Experiment 8
Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

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Title
Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application – Loan Application System.

Aim
To configure user permissions, roles, role hierarchy, and data sharing rules in the Loan Application system developed in Zoho Creator, and to study how role-based access control regulates access to application data.

Objective
To configure user management, user permissions, roles, role hierarchy, and data-sharing rules in a Zoho Creator application and understand how role-based access control can control access to application data.

Software / Tools Required
- Zoho Creator (web-based low-code application development platform)
- Loan Application system created in Zoho Creator
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

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Introduction / Theory

User Management in Zoho Creator

User management in Zoho Creator refers to the set of features used to decide who can access an application and what each person is permitted to see and do within it. Users are added to an application and are associated with permission sets (profiles) and roles. Together, these settings control access to the forms, reports, pages, and records of the application. In the Settings area of an application, the User Permissions section groups these controls under three tabs: **Permissions**, **Roles**, and **Data Sharing**.

Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is an access-control model in which access rights are assigned to roles instead of to individual users, and users acquire rights through the roles they hold. RBAC simplifies administration, supports the principle of least privilege (users receive only the access necessary for their duties), improves data security, and keeps the access policy consistent and easy to audit.

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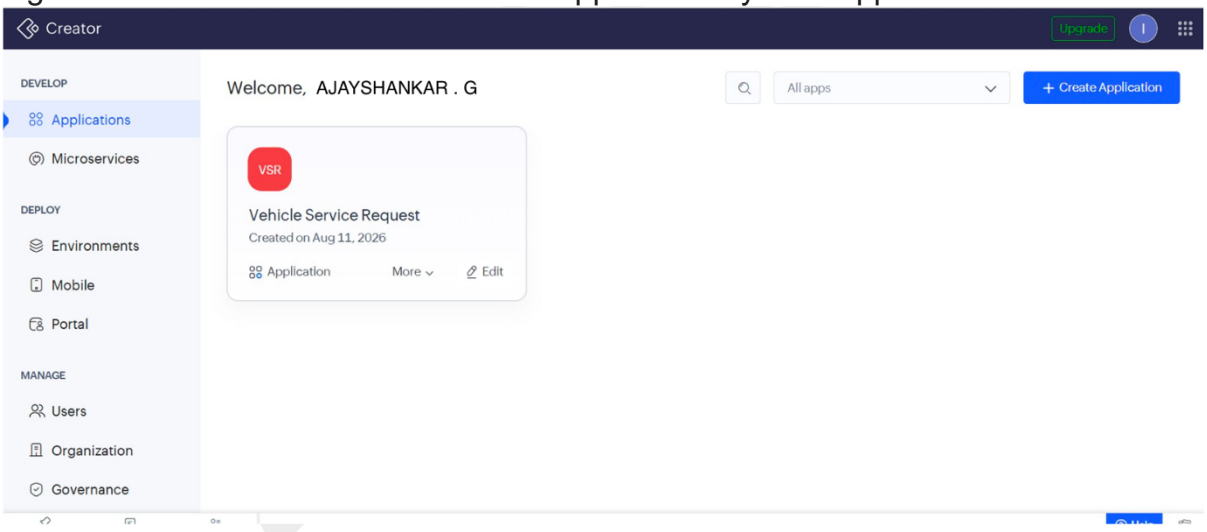
Key Concepts

Concept	Description
Users	Individuals who are added to the application and who sign in to work with its forms, reports, and data.
Permissions	Named permission sets (profiles) that define what a user may do with application components, for example Read or Write.
Roles	Positions in the organizational structure of the application (for example CEO). A role determines the position of a user in the hierarchy and the visibility of records.
Role Hierarchy	A tree structure in which each role reports to another role. A role higher in the hierarchy can generally view the records of the roles that report to it, whereas the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the default hierarchy by sharing the records of a form from one role or user to another role or user with a defined level of access (for example Read Only or Read/Write).

Procedure

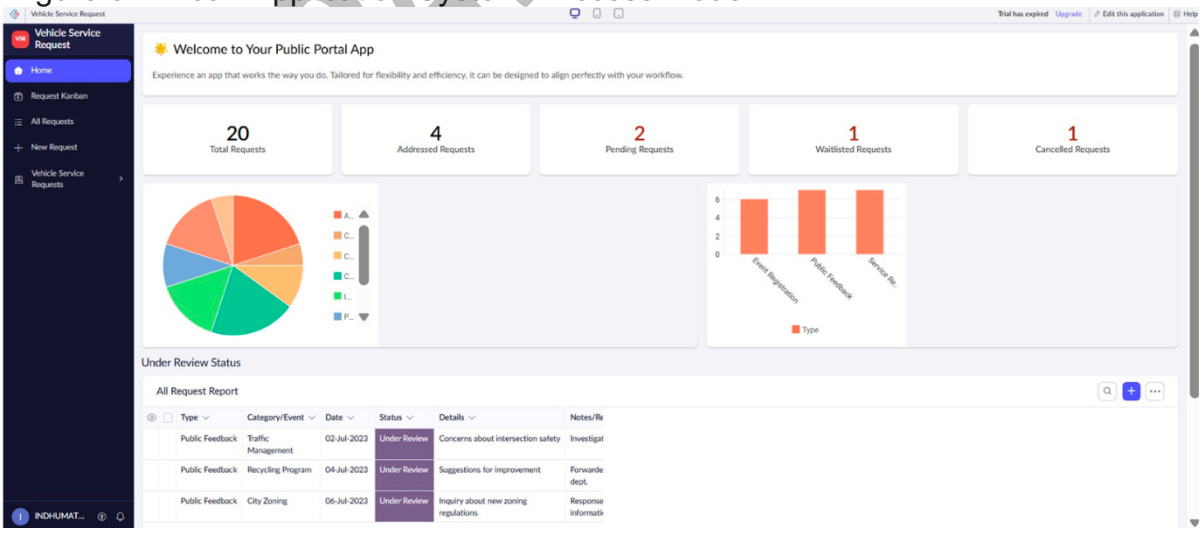
1. Sign in to Zoho Creator. The home page lists the applications available in the account. Locate the **Loan Application System** application card.

Figure 8.1: Zoho Creator Home - Loan Application System Application



2. Open the application to view its dashboard (Home page), which contains the Loan Kanban, All Loan Applications, New Application form, summary tiles, charts, and the All Request Report.

Figure 8.2: Loan Application System - Access Mode



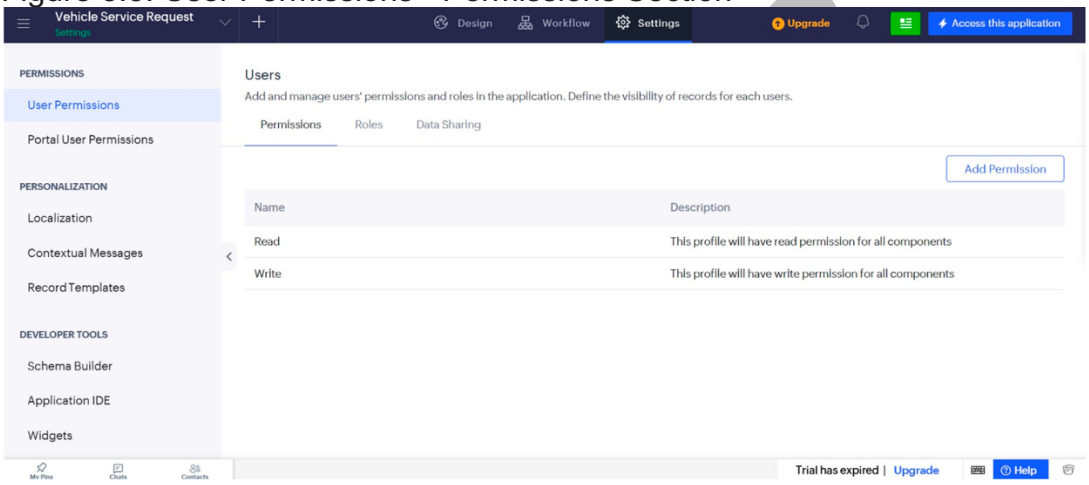
3. Enter Edit mode by selecting **Edit** on the application card or **Edit this application** at the top-right corner of the interface.

4. Navigate to Settings: In Edit mode, select the **Settings** tab from the top navigation bar (**Design**, **Workflow**, **Settings**).

5. Open User Permissions: Under the **PERMISSIONS** group of the left-hand panel, select **User Permissions**. The Users page, which is used to add and manage the permissions and roles of users and to define record visibility, is displayed with the tabs **Permissions**, **Roles**, and **Data Sharing**.

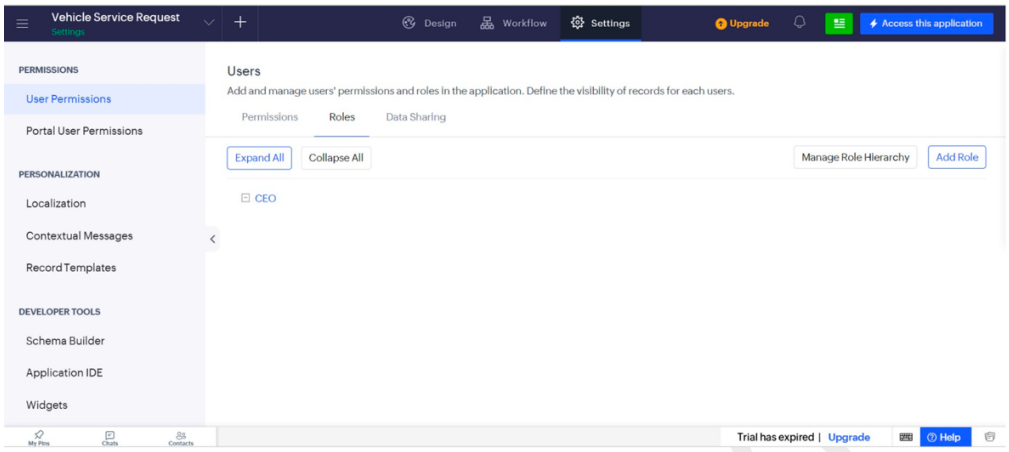
6. Study the Permissions tab: The **Permissions** tab lists the available permission sets. Two default permission sets are available: **Read** (read permission for all components) and **Write** (write permission for all components). An **Add Permission** button is also provided.

Figure 8.3: User Permissions - Permissions Section



7. Study the Roles tab: Select the **Roles** tab. The default role, **CEO**, is displayed at the top of the role tree. The **Expand All**, **Collapse All**, **Manage Role Hierarchy**, and **Add Role** options are available. The Role Hierarchy interface is used to arrange roles in a reporting structure.

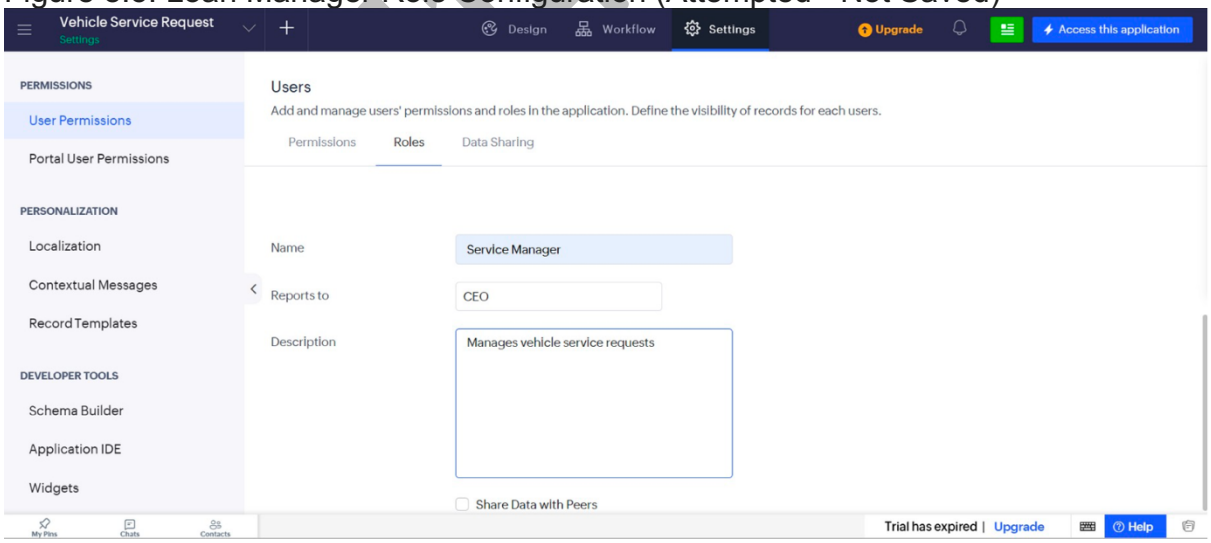
[IMAGE PLACEHOLDER 8.4: Figure 8.4: Roles and Role Hierarchy]



8. Attempt to configure an additional role: Select **Add Role** and enter the details of the role to be created:

    - **Name:** Loan Manager
    - **Reports to:** CEO
    - **Description:** Manages and processes loan applications
    - The **Share Data with Peers** option remains unselected.

Figure 8.5: Loan Manager Role Configuration (Attempted - Not Saved)



9. **Upgrade restriction:** On attempting to save the Loan Manager role, Zoho Creator displayed an upgrade request dialog. The application indicates the status *"Trial has expired"* with an **Upgrade** option, and the current plan does not permit the creation of an additional custom role. Consequently, the Loan

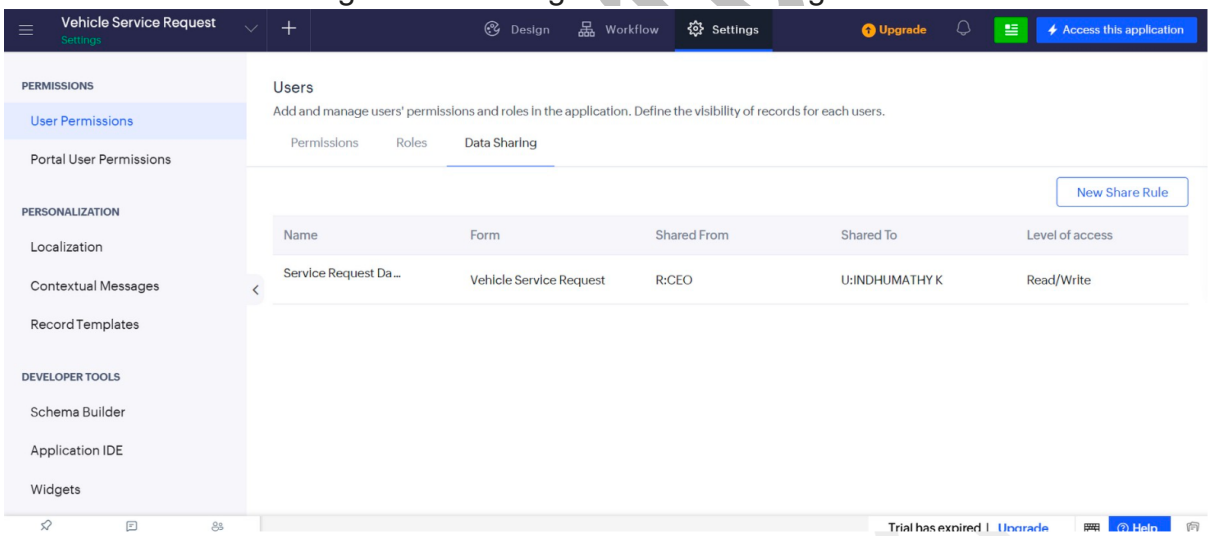
Manager role was not saved, and **CEO** remains the only active role in the application.

10. Configure the Data Sharing Rule: Select the **Data Sharing** tab and choose **New Share Rule**. In the **New Share Rule** panel, enter the rule details:

    - **Name:** Loan Application Data Sharing Rule
    - **Form:** Loan Application
    - **Records shared from:** R:CEO
    - **Records shared to:** U:[Student Name]
    - **Level of access:** Read/Write

11. Save the Data Sharing Rule: Select **Save**. The rule is successfully compiled and added to the Data Sharing list table.

Figure 8.6: Configured Data Sharing Rule



RBAC Working

In a role-based design, users are placed into roles, and the roles are arranged in a hierarchy according to the organizational structure. The intended structure for the Loan Application System is a two-level hierarchy:

Level	Role	Intended Function
1 (Top)	**CEO**	Overall authority; able to view the records of roles reporting to it.
2	**Loan Manager**	Manages loan applications; reports to the CEO.

Note on Roles actually available in the application: Only the default **CEO** role exists in the application. The **Loan Manager** role was configured only up to the point of an attempted save and was not created because the current trial plan does not allow an additional role. Accordingly, Level 2 in the table above represents the intended structure only, and no additional users were assigned to separate custom roles.

Conceptually, once the **Loan Manager** role is unlocked and available, the hierarchy operates as follows:

- Users assigned to the **Loan Manager** role work with loan application records according to their assigned permission set (Read or Write).
- Because the **Loan Manager** role reports to the **CEO**, the **CEO** role, being higher in the hierarchy, can implicitly view all records belonging to the **Loan Manager** role.
- Users in the lower role do not automatically gain access to the records of the higher role unless a specific data-sharing rule is configured.
- Permission sets (Read or Write) further restrict or grant specific operations that each user may perform on application components.

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Data Sharing Analysis

Role hierarchy alone provides only vertical visibility of records down the reporting tree. Data sharing rules are used to grant additional lateral or custom access across roles or individual users. In this experiment, a share rule was configured for the **Loan Application** form with the following mapping:

$$\text{CEO (R:CEO)} \xrightarrow{\text{Read/Write Access}} \text{[Student Name] (U:[Student Name])}$$

- **Source of Sharing:** The records of the Loan Application form belonging to the **CEO** role are the source (Shared From: R:CEO).
- **Recipient:** The user **[Student Name]** is the designated recipient (Shared To: U:[Student Name]).

- **Level of Access:** Configured to Read/Write, allowing the recipient to both view and edit shared loan records.

The saved rule appears in the Data Sharing list (Figure 8.6), demonstrating that record visibility can be expanded beyond standard hierarchical rules in a secure and controlled manner.

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Result

The following configurations were successfully executed in the Loan Application System:

- The **User Permissions** section was accessed via **Settings**, and the default permission sets (Read and Write) were evaluated.
- The default **CEO** role and the **Role Hierarchy** management interface were thoroughly explored.
- A **Data Sharing Rule** was successfully created and saved for the **Loan Application** form, sharing records from the **CEO** role (R:CEO) to the user **[Student Name]** (U:[Student Name]) with Read/Write access.
- The creation of an additional custom role (**Loan Manager**, reporting to **CEO**) was attempted; however, saving was restricted by Zoho Creator plan limits ("Trial has expired"), leaving **CEO** as the primary active role.

Conclusion:

The experiment demonstrated the core user-management capabilities of Zoho Creator—specifically permissions, roles, role hierarchy, and data sharing—and how they assemble a robust Role-Based Access Control (RBAC) framework. The Read and Write permission profiles, the root **CEO** role, and the Role Hierarchy builder were examined, and a functional Data Sharing Rule granting Read/Write privileges on loan records was configured successfully. Although custom role creation was

bounded by account plan restrictions, the conceptual principles of hierarchical data inheritance and horizontal data sharing were validated. The exercise confirms that application security and record visibility can be governed predictably and securely using RBAC principles.

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